

Integrated design of photovoltaic power generation plant with pumped hydro storage system and irrigation facility at the Uhuele-Amoncha African community

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ARTICLE INFO

Keywords:
community
Irrigation
Photovoltaic
Pumped
Hydro
Storage

ABSTRACT

— Seasonal and location dependence of renewable energy resources have limited their applications in power generation. Energy storage systems are promising solutions to the intermittence of renewable energy resources. Rural electricity grids are faced with economic sustainability challenges due to low power demand and poverty. A rural grid design around economic drivers like agriculture and micro industries can mitigate poverty and improve economic sustainability of rural grids. This paper presents an integrated design for photovoltaic power generation with a pumped hydro storage system for irrigation and community utilization. The design explored the natural availability of water body in an elevated settlement area that offers a natural storage height for hydro energy storage. A photovoltaic generation plant was designed to power a pump as a turbine system for water storage and generation. HOMER® energy simulation software was deployed in the simulation. The result shows a satisfactory net present cost for the possible integration of a pumped hydro storage system in a photovoltaic generation plant as the most viable option to provide power at a power supply probability of 99.9% and water for irrigation. The benefits from the irrigation and water supply will enhance the community's socio-economic development and sustain the investment.

1. Introduction

Energy has evolved with time as a significant driver of industrialization. Environmental concerns due to increasing global warming and sustainability issues around fossil fuels have made renewable energy systems very attractive solutions for world energy challenges. Renewable energy sources are known for seasonal and location dependence with the resulting intermittence. Their applications are limited by their intermittent nature, which requires energy storage systems [1]. Research on suitable solutions to the problems of intermittence associated with renewable energy sources has recognized energy storage systems as a viable solution. However, there is no perfect energy storage system as its application depends on many factors. Pumped hydro storage, batteries, and fuel cell technologies have been identified as suitable energy storage systems for managing the intermittence characteristics of renewable energy sources [2].

Communities in remote areas are often not supplied by the electric grid. The lack of energy and the distance to urban centers means that the

price of candles, diesel, gas, coal, and batteries is much higher than in cities [3]. This feature imposes economic burden on the inhabitants of these communities.

The access to electricity in rural communities is related to people's standard of living and socioeconomic development. Reference [4] evaluated the role of electricity in improving people's lives. The study used the capability approach in relating the case of rural electrification in India. The research results show that access to electricity is an essential social amenity that influences people's standard of living. The study, however, noted that the benefits of electricity in the community are not evenly distributed in the sampled families. Access to electricity is positively correlated with improved security, recreational activities, adaptation to effects of climate change, and health. The author argued that access to electricity should be used to enhance social and political settings to help the community to transform electricity access to an improved standard of living. As a result, this study will endeavor to develop an electricity generation design that can be implemented in isolated riverline communities to provide electricity access. In addition, the design will include an irrigation facility to encourage food

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<https://doi.org/10.1016/j.renene.2022.08.059>

Received 31 May 2022; Received in revised form 6 August 2022; Accepted 12 August 2022

Available online 18 August 2022

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Nomenclature

AS-PSH	Adjustable Speed Pumped Storage Hydro
COE	Cost of Energy
DC	Direct Current
DERs	Distributed Energy Resources
ESS	Energy Storage System
HOMER®	Hybrid Optimization of Multiple Energy Resources
NPC	Net Present Cost
PAT	Pump as Turbine
PH	Pumped Hydro
PHES	Pumped Hydro Energy Storage
PV	Photovoltaic
RES	Renewable Energy Sources
SOC	State of Charge

production for improved living, thus providing for the community's important needs.

Considering the energy cost in Kenyan households, the work in Ref. [5] assessed the impact of solar home systems and energy-related expenditures. Using a pipeline comparison approach in more than one thousand households, it was shown that access to solar home systems resulted in increased usage of LED lamps with reduced use of incandescent lamps. There was also a reduction in the monthly use of kerosene for lighting. Kerosene deserves special attention since it may be a source of fire and home accidents. In addition, the cost of charging mobile phones was reduced with the cumulative effect of reduced energy cost. The present study aims to provide an energy alternative to a rural community to address the high energy price.

Electric systems can be considered social-technological systems, as the availability of electricity goes beyond its physical presence in locations. Using electricity improves water consumption, food, health, and education. Therefore, electricity availability in a disconnected community should be the starting point for addressing poverty and social exclusion. Electricity access is so critical that it is already well regarded as a basic human right.

Renewable Energy Sources (RES) are found in several parts of the world with zero cost and can generate sustainable electricity. For this reason, RES penetration level is very high, and it is often the only energy solution in some regions. However, RES have an undesirable characteristic of intermittence, making the use of an energy storage system (ESS) essential. Therefore, the choice of ESS should be made based on storage capacity, available budget, technical characteristics (e.g., lifespan, efficiency), and natural aspects, such as the natural water availability in the Uhuelele-Amoncha community.

According to Ref. [6], some factors can contribute to the implementation failure of an electric system in rural areas; the leading factor is the inherent equipment failure in any system. Failures and intermittency in the long term can result in a community lack of interest in the electricity project, which in the future may be a reason for abandonment. Therefore, the proposal for a reliable, robust, and low maintenance system must be considered. Deploying of renewable energy systems with low-cost storage becomes a preferable solution for accessing electricity in remote communities. Implementing a photovoltaic system combined with the pumping and storing hydraulic energy involves a significant investment, even in micro hydro-generators. Much of this cost is in the excavation and waterproofing of the reservoirs [7]. Another prominent cost is the pumping system and electro-hydraulic power generation, typically independent devices.

To Aim at greater efficiency in potential energy storage, places with natural heights over short horizontal distances will be preferred. An important factor to be considered is the presence of natural lakes or water bodies that can be used as they are or even expanded to maximize

energy storage.

In this regard, the proposed design in this study has to do with siting a photovoltaic power generation plant in Uhuelele-Amoncha community. This community is located on 5° 50.3'N latitude, 7° 55.6'E longitude along Enyum river in southeast Nigeria, which empties into the Atlantic Ocean. The major occupations of residents are fishing and subsistence crop farming. The electric energy need of the community is principally for lighting, food preservation and charging of mobile phones. The average power rating of a mobile phone charger is about 5 W [8]. There are approximately 250 persons in the community with about 50 households and an average of 5 persons per household. The total community load is around 15.25 kW.

A limiting factor of pumped hydro storage system is the electromechanical energy conversion set and the equipment needed to regulate the system's speed. Thinking about simplifying the operation and reducing the inherent costs of the system, the Pump as Turbine (PAT) concept will be proposed, which are electromechanical devices used in micro and nano hydro-generators. Due to their low cost and high reliability, PATs are great alternatives in small communities isolated from the grid as they can serve as both pump and generator.

Typical rural electricity demand is usually low due to the absence of industries, which does not encourage investments in rural electric grids. Another challenge of rural electricity investment is low payment compliance by consumers due to poverty. These result in the problem of economic sustainability as only very few rural minigrids are functional [9]. As a solution to these challenges, rural agriculture and agro-allied industries could transform the rural electricity demand and improve the incomes of rural dwellers, thereby increasing their ability to pay for consumed electricity. This is the main idea behind this design, as the research community does not widely address this problem.

This research provides a technical design to provide feasibility details of applying pumped hydro storage in photovoltaic plants and application in agricultural irrigation. In summary, this paper contributes to an integrated design for the productive use of electricity in rural agriculture with the following benefits:

- Energy access to a remote agrarian community;
- Water for irrigation for improved crop yield and food security;
- Water supply for improved community hygiene;
- Encourage environmental sustainability through renewable photovoltaic energy generation and pumped hydro storage system;
- Ultimately guarantee economic sustainability of the rural microgrid through a design around agriculture as an economic driver.

The paper is organized as follows: Section 2 presents a review of related literature, Section 3 is design specifications and simulations, Section 4 presents results and discussion, while Section 5 contains the conclusion.

2. Review of related literature

Various scholars have provided solutions to the intermittence problem of renewable energy generation. Tilahun [10] proposed a solution to control generation capacity fluctuation due to seasonal water level variations using a pumped hydro storage system for Tana Beles hydropower. The author proposed a pumped storage system with six reversible pumps/turbines. The expected benefits of the pumped storage system will include cost reduction and power availability for peak hour power demand.

Manolakis et al. [11] implemented a stand-alone photovoltaic plant that uses hybrid storage of batteries and a pumped hydro storage system on Donoussa Island. The total installed photovoltaic generation capacity was 18 kWp. They used a pump rated 6 kVA and a water turbine coupled to a 7.5 kW Direct Current (DC) generator. Two identical reservoirs of 150 m³ capacity each were used in the lower and upper storage. The daytime energy demand of the village was met through direct supply

Table 1
Summary of related literatures.

Ref.	Country	Assumed resources ^a						Optimal LCOE (USD/kWh)
		Solar	Batteries	Biomass/Biogas/biodiesel	Wind	Hydro	Diesel	
Lemence et al. [27]	Philippines	✓	✓				✓	0.54
Ribó-Pérez et al. [28]	Honduras and Zambia	✓	✓	✓				0.06
Laflleur et al. [29]	Canada						✓	No info
Chambon et al. [30]	India	✓	✓	✓			✓	0.18
Odou et al. [31]	Benin	✓	✓			✓	✓	0.207
Atallah et al. [32]	Egypt	✓	✓		✓		✓	0.107
Lozano et al. [33]	Philippines	✓	✓				✓	No info
Aziz et al. [34]	Iraq	✓	✓			✓	✓	0.054
Muh et al. [35]	Cameroon	✓	✓		✓	✓	✓	0.443
Haratian et al. [36]	Iran	✓	✓		✓			0.546
Azimoh et al. [37]	Namibia	✓	✓				✓	0.51
Ali et al. [38]	Australia	✓	✓		✓		✓	0.238
Shahzad et al. [39]	Pakistan	✓	✓	✓				0.026
Ramchandran et al. [40]	India	✓	✓	✓				0.23
Mamaghani et al. [41]	Colombia	✓	✓		✓		✓	0.465
Chmiel et al. [42]	Scotland	✓	✓		✓	✓	✓	0.238
Bhattacharyya [43]	Bangladesh	✓	✓		✓		✓	0.344
Sen et al. [44]	India	✓	✓	✓	✓	✓		0.42
Kim et al. [45]	South Korea	✓	✓		✓		✓	0.23
Dorji et al. [46]	Bhutan	✓	✓		✓		✓	0.582
Prodromidis et al. [47]	Greece	✓	✓		✓			0.42
Prodromidis et al. [48]	Greece	✓	✓		✓			No info
Nfah et al. [49]	Cameroon	✓	✓	✓		✓		0.352
Nfah et al. [50]	Cameroon	✓	✓			✓	✓	0.296
Average								0.297

from the photovoltaic generator using a 25 kVA, 390 V, 3-phase inverter. Surplus energy is used to power the pump to fill the upper reservoir. During the night, water is released from the upper reservoir to drive the turbine for power generation to supply the loads. The battery bank with 100 Ah capacity provides primary load peaks. The authors, however, noted that an isolated photovoltaic generation plant like this has a high initial cost that may deter investments if there are no other benefits like water supply and irrigation.

Hence, the present study will attempt to provide a design that will cater for a community's power generation, water supply, and irrigation needs on a hill along a river.

In a recent development, Obermeyer [12] proposed a cost-effective small-scale adjustable speed pumped storage hydro (AS-PSH) system optimized for the United States' energy storage needs. The application was proven through concept design for selected sites with associated costs. The project shows that the proposed technology is commercially viable and can economically serve the energy storage needs of the identified markets with optimal installation opportunities. This report lays credence to the economic applicability of small hydro storage systems in the energy mix of any country with supporting natural water supplies and storage heights. Uhuelel-Amoncha community has a natural water supply through Enyum river and a natural elevation above sea level that can support the location of pumped hydro storage.

To establish the economic viability of a pumped hydro storage system, reference [13] investigated the benefits of using pumped-storage hydropower in modern power systems with increasing RES generation in a liberalized energy market. The authors developed a novel operation algorithm that balances providing additional capacity to meet peak load and obtaining a justified financial revenue for the high investment cost in a pumped hydro energy storage system. The proposed algorithm was implemented using MATLAB® to analyze the benefits of the Tonstad pumped hydro energy storage system. The results showed that the storage could provide significant revenue in the liberalized electricity market of Norway while reducing fluctuations in the overall system loads.

Reference [14] proposes an optimal design/operation scheme for residential neighborhoods in the context of distributed energy resources (DERs) integration. The overall conclusion is that DERs outperform the conventional grid regarding CO₂ emissions and cost.

Reference [15] focuses on irrigation efficiency, which is linked to this paper's proposal of providing reliable irrigation for farming purposes. The proposed scheme in Ref. [15] is highly effective as it can reduce water demand by up to 54% due to increased efficiency.

Reference [16] focuses more on social aspects as remote villages in Cameroon were assessed. The idea is similar to this paper, i.e., to propose hybrid PV/hydro systems. Reference [16] also applies the software HOMER® due to its wide recognition and ability to utilize geographic data to arrive at optimized solutions peculiar to a design location. While the idea of [16] and this paper are similar, different remote villages inherently lead to distinct results/design specifications. Therefore, publications on the topic contribute concurrently to mitigating energy poverty in several communities worldwide.

Reference [17] conducted a thorough literature review on hybrid systems in remote communities; thus, it is critical to understand the trends and characteristics of the proposed solutions. According to Ref. [17], more than 90% of studies are conducted in the northern hemisphere, which is a concern. Furthermore, more than 60% of studies combine PV and wind power, and batteries are preferred as the energy storage devices in more than 80% of studies. Reference [16] demonstrates that while PH is a promising solution, it is hardly ever proposed since it requires natural water availability. Therefore, it is fair to affirm that the solution proposed in this paper goes against natural trends in hybrid systems in remote communities.

Reference [18] focuses on optimizing hybrid energy systems in Nigeria; however, neither hydro generation nor HOMER® application is addressed.

Reference [19] argues that wind-diesel-battery systems tend to be superior in the isolated community of French Island (Victoria, Australia). However, as previously mentioned, the feasibility of PH depends on natural water availability. Furthermore, it is widely recognized that diesel generation leads to adverse environmental impacts [20].

Reference [21] focuses on a different approach as hybrid wind and tidal generation is proposed in a remote community in New Zealand. The main conclusion is that it is possible to decrease CO₂ emissions and net present costs (NPC) based on such a design.

Reference [22] focuses on hybrid diesel-PV systems in remote communities in Afghanistan to decrease cost and mitigate energy poverty. Community-based hybrid energy systems are further assessed by Ashok

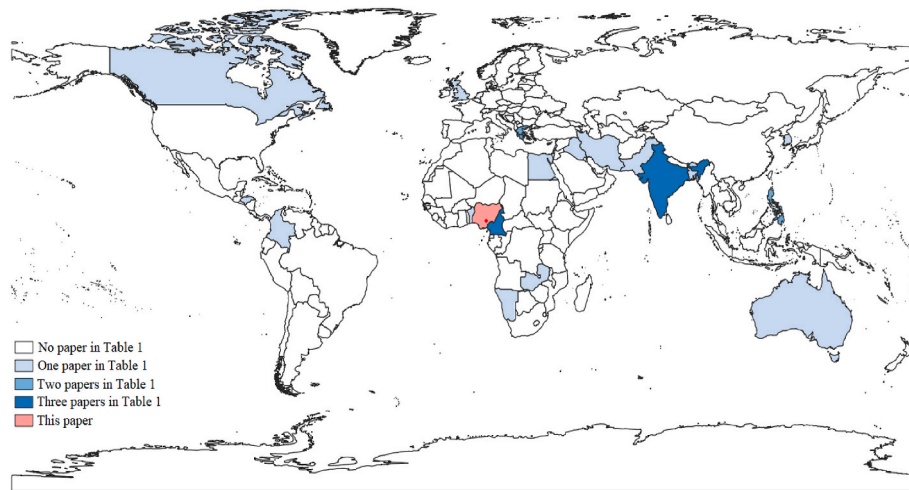


Fig. 1. Map of high-impact Renewable Energy publications on the topic (exact location addressed in this paper is highlighted with a red dot).

[23] who opined that hybrid energy system is an excellent solution for rural electrification in remote communities with grid extension challenges and economic drawbacks. Ma et al. [24] proposed a hybrid distributed energy resource (DER) system integrating combined heating and power (CHP), photovoltaic (PV) and wind power as solutions to the electrical and thermal load requirements in community residential buildings. Integration of storage systems in microgrids and conventional distribution systems are viable solutions to the intermittence of renewable generation [25]. Hybrid energy resources of wind and hydro energy have the potential to provide Ireland with energy security at a minimal emission level [26].

For a thorough literature review related to the context of this paper, the following descriptors were applied to the Scopus database: “HOMER” AND “hybrid” AND “off-grid”. It was verified that Renewable Energy is the leading journal on the topic with 24 high-impact publications, as summarized in Table 1. A significant diversity of countries is demonstrated. For better visualization, Fig. 1, illustrates a map of the publications. This paper is the only one out of the 24 that addresses an isolated Nigerian community, which is a country that suffers from energy poverty and deserves greater attention from the research community. Concerning resources, a significant variety was also verified, with several papers assessing electricity generation from solar, biomass, wind, hydro, and/or diesel, along with battery energy storage. However, none of the 24 papers addressed pumped hydro storage systems (a major focus on batteries is evident). In conclusion, this paper focuses on an understudied country and system; thus, it is expected to be of great value for the research community.

Also, among high-impact publications closely related to this paper's context, stand out: Dumbrava et al. [51] focus on PV production management and stochastic microgrid optimization. The assumed microgrid is robust with PV generation, thermal engine, ESSs, and critical and interruptible loads. Moreover, it can operate in islanded mode. The goal is to stochastically minimize operation costs, which was achieved by the proposed model. Lazaroiu et al. [52] addressed mathematical formulation and implementation of virtual power plant (VPP). The scheme assumes renewables, ESSs, thermal engines, and demand-response loads, and seeks to maximize the VPP's benefits. The model is implemented based on the GAMS software and results demonstrate that it successfully maximizes benefits.

This study will propose an integrated off-grid pumped hydro energy storage (PHES) powered by a photovoltaic (PV) plant for irrigation and household consumption in an isolated community. The goal is to design a durable system with low maintenance and ease of operation.

Solar energy has been identified as the most abundant energy resource on Earth. The amount of energy radiating from the Sun to the

Earth matches the energy demand by human activities on Earth. PV technology is among the major means of harvesting energy from the Sun, and many technological advances aim at improving the efficiency of photovoltaic systems. PV technology is a mature and commercially available technology with proven reliability [53]. Due to the environmental sustainability characteristics of PV generation, many nations have made policies to encourage investments in PV generation. This is evident in incentives, subsidies, and tax exemptions granted to investors in PV technologies [54].

In a system where photovoltaic solar energy is expected to supply water needs for irrigation and human consumption, the PV plant needs to be well sized, so that the generated electricity under normal conditions, is sufficient to pump and store water in the upper reservoir and supply energy to the homes in the community. Another important point is the quantity of electrical power needed for the houses during the day. To predict the households loads behavior, the HOMER® software is adopted to provide a generation estimate in the proposed model. From the data analysis, an electric system powered by photovoltaic panels will be planned. Hence it is expected that the system should be able to supply all the electrical power demand and water pumping as a means of energy storage and community usage at the same time.

2.1. Energy storage system

When considering isolated low-income communities, it is necessary to think about a long-lasting, reliable, low-maintenance energy storage system. Battery storage systems are the most versatile in electrical conversion. However, they are not feasible in many parts of the world due to cost and sustainability. The same can be said about fuel cells that require special care in fuel storage and their high cost. The option adopted in this research is hydraulic storage for several reasons. First, the feasibility of such a system is linked to the availability of natural slopes where reservoirs can be created. This premise is important due to the high costs associated with excavating reservoirs [7]. The storage capacity of the reservoirs will meet the community's water needs for human consumption, energy generation, irrigation, and animal use. In this holistic view, the storage system fulfills much more than the energy function but meets other basic human needs.

2.2. Pump as turbine

Pump as turbine (PAT) is not necessarily a new concept, but its use is not yet widespread. Arriaga in his work [3], proposes that an isolated community can have PAT as an alternative for energy supply. This proposal is interesting due to the good energy quality and cost-effective

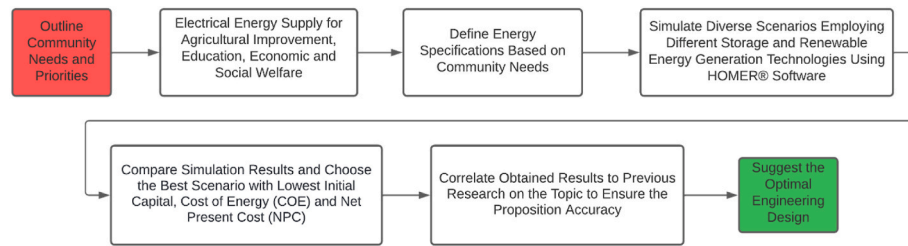


Fig. 2. Procedure flowchart.

Table 2
Rural household load specifications [55].

Load	Quantity	Power [W]	Total Power [W]	Day Time
LED light bulb	5	10	50	05:00–06:00 h 18:00–22:00 h
Smartphone charger	3	5	15	08:00–10:00 h 18:00–20:00 h
Fan	1	60	60	19:00–23:00 h
Fridge	1	40	40	00:00–23:00 h
26" LED TV	1	80	80	17:00–21:00 h
Laptop	1	60	60	18:00–22:00 h

results for rural electrification, and these facilities can be installed and maintained by the villagers. Indeed, using a pump-type system as a turbine is one of the best solutions that can be applied in remote communities. This topology allows the use of the same hydraulic turbine, electric machine and piping for water pumping and electricity generation, significantly reducing the implementation cost [3].

2.3. Community irrigation application

With respect to irrigation application, since the community largely depends on subsistence crop farming for a livelihood, incorporating an irrigation facility in the design will enhance food production and improve the community's economy. In this regard, power generation will be prioritized using stored hydro potential energy during renewable energy downtime. The excess of accumulated water in the storage system will be used for irrigation purposes.

2.4. Generation and load controller

The design will employ “HOMER® Cycle Charging” controller to ensure good energy management in the reservoir's loading and unloading cycles.

3. Design specifications and simulations

To carry out the dimensioning of the hybrid photovoltaic-hydraulic system, it is necessary to know precisely the loads connected to the electric microgrid and the amount of water used in agriculture so that the community does not lack water or energy. This information is valuable because it will also serve as input to the HOMER® program to help the system simulation determine which specifications will be the most viable.

The procedure flowchart for the study is shown in Fig. 2. This presents the processes involved in the design.

3.1. Load specifications

Quantifying the electricity consumed by the community during the day is the first step in planning the photovoltaic system. To assist in this dimensioning, the model of a single-family residence of five people will be proposed, focused on the real necessities of the community under

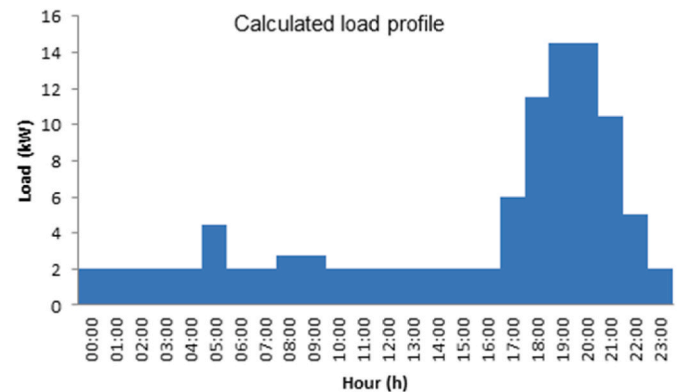


Fig. 3. Calculated load profile.

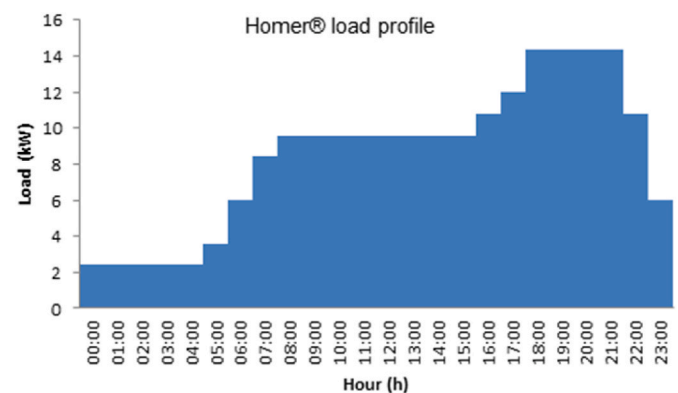


Fig. 4. HOMER® Load profile.

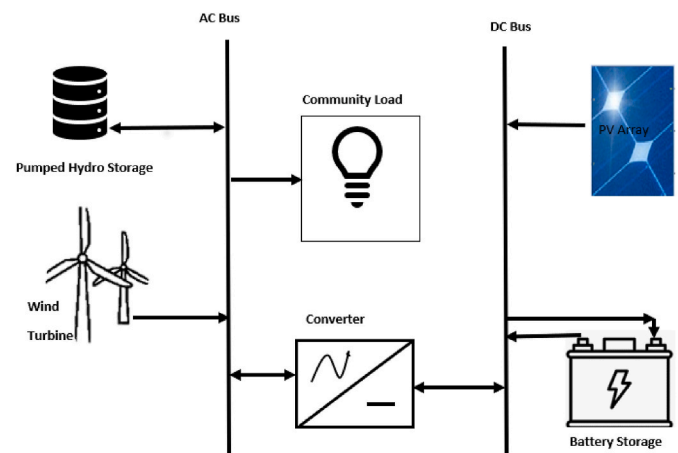
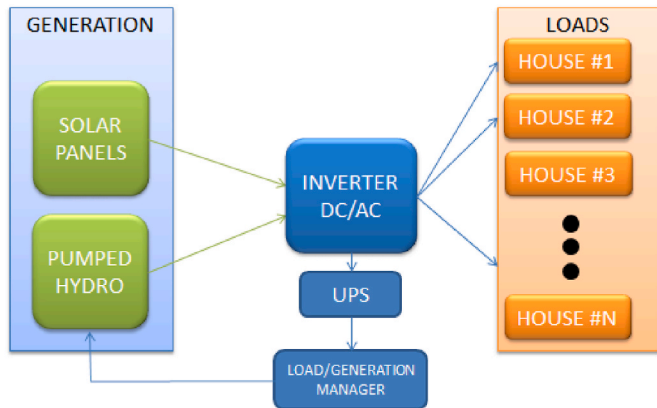


Fig. 5. Hybrid PV-PH schematic.

Table 3

Architecture scenarios comparison based on average 165 (kWh/day) consumption.

System Architecture					Initial Capital (US\$)	NPC (US \$)	COE (US \$/kWh)	Operating Cost (US \$/yr)
PV (kW)	10 (kW) Wind Turbine (un.)	245 (kWh) PH Storage (un.)	100 (kWh) Li-Ion Battery (un.)	Converter (kW)				
98.4	0	2	0	20.8	88,071	194,385	0.129	4253
67.6	1	2	0	19.7	125,738	256,750	0.170	5240
138	1	0	3	21.3	319,738	563,715	0.374	9759
148	0	0	4	21.6	343,491	634,230	0.421	11,630

**Fig. 6.** Scheme of the components involved in the off-grid hybrid PV-PHES system.

study; its load and power consumption follows the distribution presented in Table 2.

Analyzing the information in Table 2, one can see that most of the energy consumption occurs between 17:00 and 22:00: By multiplying the values in Table 2 by 50 houses, the community load curve shown in Fig. 3 is obtained from the data in Table 2. A comparison was made between the calculated curve and the “community” load curve provided by the HOMER® software shown in Fig. 4.

Different scenarios were simulated using the load and location data to determine the best system based on cost and economic sustainability. Fig. 5 shows the various generation scenarios simulated while Table 3 shows their respective costs.

The proposed scheme is depicted in Fig. 6. The first scenario in Table 3 was chosen for implementation because it offered the lowest cost of energy to meet the load demand and provide water for irrigation through the use of pumped hydro storage.

The HOMER® software presents a higher load curve, requiring a more robust generation system. For this reason, the system specifications will be based on the load curve provided by HOMER® because its dimensions exceed the calculated minimum necessary. This is particularly important since the load profile in Fig. 3 is an estimation and not an actual measurement, implying a security margin is justifiable, and the energy required to power the PAT in pump mode. Therefore, it will give a greater operating margin for the system, with a higher probability of meeting the irrigation application needs of the people. This information shows the scaled average daily energy use is 165 kWh.

3.2. Photovoltaic system

According to Solar Radiation Database of National Renewable Energy Laboratory (NREL), the average global solar radiation at Uhuele-Amoncha, Nigeria, is 4.5 kWh/m² a day [56]. It is expected that the photovoltaic system will produce more energy than necessary for community use. It is common to have days with low solar radiation due to rain or even the presence of clouds.

The photovoltaic generator aims to meet two different demands in

the community. During the day, it is expected that there will be electricity from solar sources to store water in the reservoir and supply energy to the community. It is appropriate to separate the photovoltaic generator system into two units. The first one is to be installed closer to the houses and the second PV generator should be installed close to the pumps in order to supply the converter.

The energy produced by a PV power plant is given by Equation (1) [57]. Based on the HOMER® community load profile, it was observed that the scaled daily average energy consumption is 165 kWh. Applying Equation (1), a set of 83 typical photovoltaic 460 W panels, resulting in a 38 kWp plant to meet the load requirements.

$$E = P_n \left(\frac{A_p}{P_p} \right) R_m \eta_p \quad (1)$$

where P_n is the nominal power of the plant in kWp, A_p , P_p , and η_p refer to the area, nominal power, and yield of a typical solar panel (i.e. η_p is the efficiency of the photovoltaic panels), and R_m is the average global solar radiation.

Equation (1) defined the power plant size based on average daily solar irradiation. To supply the total electrical demand during night and cloudy days, the pumping system must pump and store enough water to provide energy to the community during these situations. This is especially important at peak consumption between 18:00 and 21:00. Defining the amount of energy to be stored in the upper reservoir is an optimization problem due to uncertain weather conditions. In this sense, cloudy periods for up to four days are expected even in a location with high solar irradiation. It is also important to remember that even on cloudy days, the solar system can generate energy with reduced capacity.

For this reason, the photovoltaic system must be at least 2.5 times larger than that theoretical value. With this information, it is possible to size the project's photovoltaic system to meet the community's energy demand and water consumption. As a result, the photovoltaic system will be approximately 95 kWp of installed power. As shown in Section 3.1, the HOMER® software was also used to perform the sizing of the PV system to establish a comparison. The power sizing of the PV system was done with data from the HOMER® Optimizer™ resulting a 98.1 kWp PV System. The same strategy was used to determine the best inverter regarding the community load. The simulation with the software included historical meteorological data, thus giving greater reliability to the installed power sizing.

3.3. PAT and hydrostorage system

3.3.1. Pump mode model of the PAT

The models adopted in this design were according to Ref. [58]. When solar radiation is available, the pump as turbine (PAT) operates as a pump storing water in the upper reservoir. The electrical energy used by the pump to store water in the upper reservoir is modelled by Equation (2):

$$E_p = \frac{\rho g h Q_p}{\eta_p} \quad (2)$$

where E_p is the electrical power delivered to the pump by the PV system,

ρ is the water density = 1000 kg/m³, g is the acceleration due to gravity 9.8 m/s², h is the height of the upper reservoir from the river, Q_p is the volumetric flow rate of water from the pump to the upper reservoir, while η_p is the pump efficiency.

3.3.2. Generator mode model of the PAT

During the absence of generation from the PV system, the PAT operates as a turbine generating electric power from the potential energy stored by the water in the upper reservoir. The electrical energy generated by the turbine is given by Equation (3).

$$E_G = \rho g h Q_t \eta_G \quad (3)$$

where E_G is the electrical energy generated by the PAT running as a turbine, ρ is the water density = 1000 kg/m³, g is the acceleration due to gravity 9.8 m/s², h is the height of the upper reservoir from the river, Q_t is the volumetric flow rate of water from the upper reservoir to the turbine, while η_G is the generator efficiency.

3.3.3. Reservoir specifications

The quantity of water in the upper reservoir at any instant of time is specified by Equation (4).

$$Q_{sw} = (1 - \sigma)Q_{sw}(t-1) + \int_{t=1}^t Q_p(t)dt - \int_{t=1}^t Q_t(t)dt \quad (4)$$

where Q_{sw} is the quantity of stored water, σ is the quantity of water lost to evaporation and leakages.

The state of charge of the upper reservoir and total energy stored in the reservoir are given by Equations (5) and (6), respectively.

$$SOC(t) = \frac{Q_{sw}(t)}{Q_{swmax}(t)} \quad (5)$$

constrained by $Q_{swmin} \leq Q_{sw} \leq Q_{swmax} = C$. Considering that:

$$E_{sw} = \rho g h C \quad (6)$$

where C is the reservoir capacity, E_{sw} is the energy stored in the upper reservoir, SOC is the state of charge of the upper reservoir.

$$C = \frac{E_{sw}}{\rho g h} \quad (7)$$

As stated by de Doile et al. [57], the NPC is given by (8), where C_0 is the initial cost, CF_i is the annual cash flow, and r is the project expected return rate in a period of n years.

$$NPC = -C_0 + \sum_{i=1}^n \frac{CF_i}{(1+r)^i} \quad (8)$$

With the pumped photovoltaic system installed, it is possible to store water for irrigation and energy purposes. This photovoltaic system will produce an average daily energy of 425 kWh, out of this value, 165 kWh is required for community load while the remainder of 260 kWh is available for energy storage in the reservoir. Considering losses and a head of 35 m, approximately 2500 m³ of water can be pumped on average daily. Another challenge is to define the exact reservoir capacity, since the size must be based on historical meteorological data to forecast rains and cloudy days. Based on available information from Section 3.1, an energy storage of 500 kWh or approximately 5000 m³ is capable of meeting the community electricity and irrigation needs. Two units of PAT rated 15 kW each were proposed for the design.

Analyzing the options available in the HOMER® library, there is the option “Generic 245 kWh Pumped Hydro” is a 1000 m³ reservoir with a head of 100 m. The Uhuelele-Amoncha location does not have this head level, and the indicated installation site has an average head of 35 m. Due to this difference, the reservoir volume was changed to 3000 m³ to maintain the same energy as the original system. The reduction of the

Table 4

System cost.

System	NPC (US \$)	COE (US \$/kWh)	Oper. Cost (US \$/year)	Initial Capital (US\$)
Solar-PH	194,314	0.129	4235	88,429

head decreases the expenses with piping, but on the other hand, the cost of preparing the upper reservoir increases; hence, such consideration is reasonable as it possibly does not significantly influence the results. Therefore, a 5% minimum state of charge (SOC) will be adopted at simulation to supply the irrigation system since such SOC proved sufficient to comply with the community's irrigation requirements.

With the photovoltaic system producing more energy than needed during the day, the excess electrical power will be used in the pumping and water storage system in the upper reservoir using the PAT system. The “Generic 245 kWh Pumped Hydro” is a 1000 m³ reservoir with a head of 100 m. Therefore, it was decided not to change the average value of US\$ 22,000 suggested by the HOMER® and the same was done with the maintenance and operation values. This is because HOMER® is a worldwide recognized software and most probably suggests reasonable cost values.

In “HOMER® Optimizer™” simulation mode, two systems of the “Generic 245 kWh Pumped Hydro” type were suggested. The energy stored by these reservoirs will be sufficient to meet the electrical power needs of the community. As a consequence, a minimum reservoir of 6000 m³ for energy storage purposes is expected. According to Ref. [59], 1 ha of pumpkin vegetable farm requires an average of 1500 m³ to 2500 m³ of water per annum. With this information, the irrigation water needed for 10 ha of pumpkin vegetable farm was arrived at as follows:

In a day, 1 ha will need 2500/365 = 6.85 m³ of water. Hence 10 ha of pumpkin vegetable will need 10 × 6.85 = 68.5 m³ per day. Approximately, 10 ha of pumpkin vegetable farm will need 70 m³ of water daily. In addition to the quantity of water required for the daily electricity generation, 70 m³ of water is needed for irrigation.

3.4. Simulation

Costs of equipment are required for simulation to assess the economic viability of the proposed design. In this case, values based on market research were used for the photovoltaic system and converter. As for the “Pumped Hydro” system, the average values provided by the HOMER® were used without changes. The working diagram of the solar/hydraulic system is shown in Fig. 6 and the proposed system architecture is specified as follows:

Control: Homer cycle charging;
 PV rating: 98.7 kW;
 Pumped Hydro Storage system: PH 245–2.00;
 DC-AC Converter: 21.8 kW.

4. Results and discussion

At the end of the simulation, the annual production of the photovoltaic system is 136,785 (kWh). The initial cost of the entire system was approximately US\$ 88,429, while the net present cost (NPC) is US\$ 194,314. Finally, the average cost of energy (COE) is US\$ 0.129/kWh.

Table 5

Solar-PH Cost Summary.

Component	Capital (US\$)	Replac. (US\$)	O&M (US\$)	Total (US\$)
PH 245 kWh	44,000.00	3000	100,000.00	146,571.43
Flat plate PV	38,498.89	0	411.31	38,910.20
Converter	5930.25	4352.47	0	8831.90
System	88,429.14	7352.47	100,411.31	194,313.52

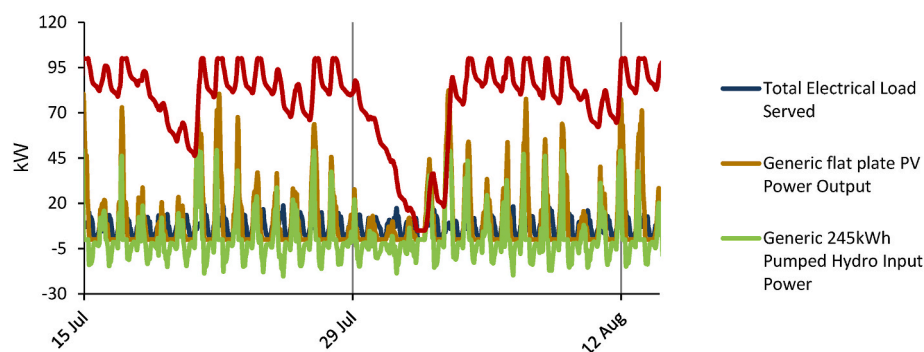


Fig. 7. Total electrical load served, PV power output, and pumped hydro input power.

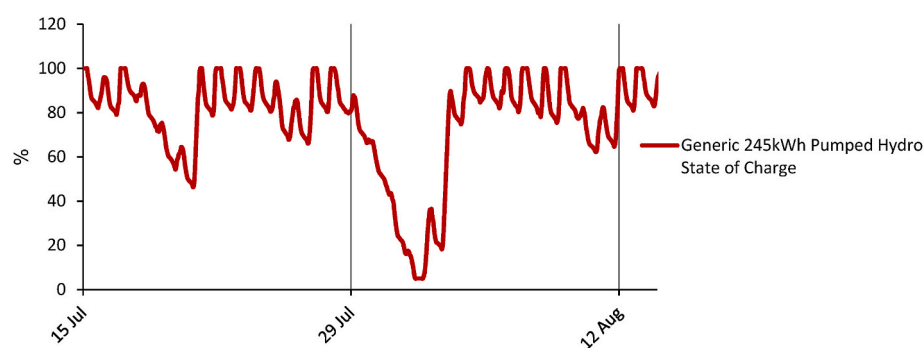


Fig. 8. Pumped hydro state of charge.

Therefore, the COE is on the same order of magnitude as previous research on the topic (Table 1), being lower than the average due to the high hydroelectric potential of the community. The average residential energy tariff in Nigeria [60] is US\$ 0.058/kWh. Simulation results for the expected 25 years lifetime of the system is shown in Table 4.

Considering Table 4, the most expensive part of the NPC is the operation and maintenance of the pumped-hydro system during its lifetime. Table 5 shows the cost summary of the system.

The “HOMER® Cycle Charging” controller operates the to ensure good energy management in the reservoir’s loading and unloading cycles. At some times, the PH system operates at positive power, consuming energy, thus indicating pumping, and at other times in negative power, meaning energy generation while functioning as a turbine.

Figs. 7 and 8 show that the pumped hydro system typically operates with a SOC of around 70–100% in days with satisfactory generation from the PV system. However, by assessing the period from July 29 to August 03, it can be verified that the SOC dropped significantly as there were three days without solar irradiation, making it necessary to consume a significant portion of the stored energy. The designed system can supply the community if there are no several days without consecutive solar irradiation. Therefore, the designed system can be considered reliable and robust under these assumptions and is expected to mitigate energy poverty issues in the Uhuelele-Amoncha region. This statement is corroborated by the loss of power supply probability of only 0.1%.

For more in-depth analysis/conclusions, a sensitivity analysis is conducted concerning the community’s electricity consumption, as illustrated in Fig. 9. This is particularly important since although reliable data is used to estimate the consumption, it might be slightly different from expected. It can be verified that the COE results are relatively robust in terms of the electricity consumption. Therefore, the system would be satisfactory even if when the consumption is varied.

5. Conclusion

This work proposed a sustainable integrated strategy to help an African community to access electrical energy and water supply for irrigation and domestic use. Based on the simulation results from HOMER® software, the scaled electrical power requirement of an Uhuelele-Amoncha community is 165 kWh/day. Therefore, the Photovoltaic generation plant modeled in this study is sized to provide the scaled power need of the community and supply the water needed for the irrigation of 10 ha of pumpkin vegetable farm. The system based on the integrated design of photovoltaic power generation plant with pumped hydro storage system and irrigation facility is considered reliable, with a power supply probability of 99.9%.

Considering the lack of matching generation, transmission, and distribution capacity against the country’s energy demand, the community in this study suffers energy exclusion from the grid power supply. Therefore, such cost comparison with standing energy tariffs will not be justifiable. Also, the proposed scheme will provide the community with irrigation facilities and water supply. The riverine community, relies on fishing activity, and electricity can be used in fish preservation for increased profit. Fish can be stored for later consumption with refrigerators, while enabling fish transportation for sale in cities.

The cost summary shows that a pumped hydro storage system attracts an operation and maintenance cost of more than 50% of the NPC. This high cost of operation and maintenance can be reduced by training some community members on specific maintenance and operation practices.

It is expected that access to electricity generates a virtuous circle, giving basic services to people in isolated communities, and promoting improvements in their living standards, thus justifying investment in the energy system. The community’s growth translates directly into the development of the region and the country. The study provides project feasibility information on the possible application of the proposed design in energy disadvantaged communities. Reference [16] also applies the software HOMER® due to its wide recognition. While the idea

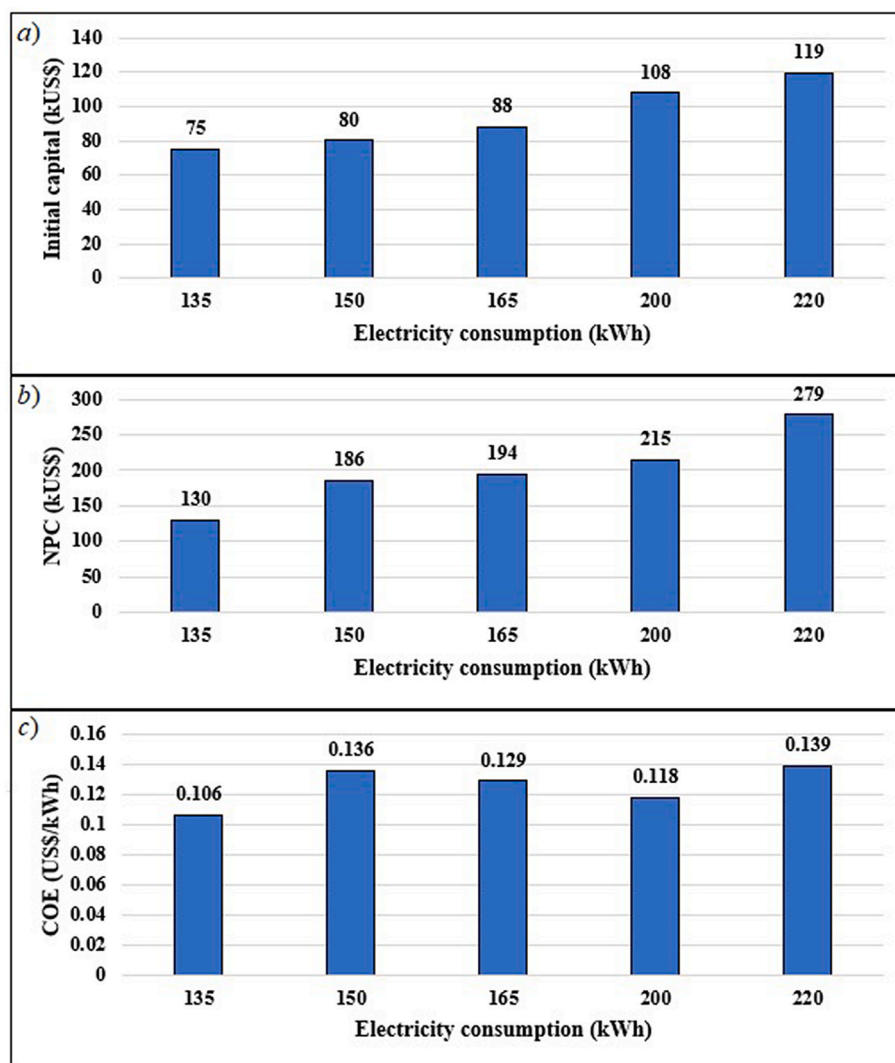


Fig. 9. Sensitivity analysis a) Initial capital, b) NPC, and c) COE.

of [16] and this paper are similar, different remote villages inherently lead to distinct results/design specifications. Therefore, publications on the topic contribute concurrently to mitigating energy poverty in several communities worldwide.

The problem of economic sustainability challenges in rural electricity grid due to low power demand and poverty is targeted and can be addressed through microgrid designs around rural agriculture and agro-allied industries to transform the rural electricity demand and improve the incomes of rural dwellers, thereby increasing their ability to pay for consumed electricity.

As a possible solution to energy shortages in rural communities, utility companies can partner with the government, international donor agencies, research organizations like Forum for Agricultural Research in Africa (FARA), and the community to finance, execute and manage the project for an improved standard of living in the affected communities. It is important to note that HOMER® does not calculate/estimate social benefits. Hence the social benefits derivable from the irrigation and water supply component of the design is a subject of another socio-economic research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Acknowledgment

The authors gratefully acknowledge the financial support in part of CAPES - Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brazil - Finance Code 001, CNPq - National Council for Scientific and Technological Development - Brazil, INERGE, FAPEMIG - Fundação de Amparo à Pesquisa do Estado de Minas Gerais, and UNIFEI. Akanu Ibiam Federal Polytechnic Unwana, Nigeria, Tertiary Education Trust Fund (TETFund), Nigeria under the leadership of the executive secretary Prof. S.E. Bogoro; the chairman, Steering Committee for Implementation of FARA-TETFund-UFV Pedagogic Retooling Strategy, Dr. Mustapha Popoola and Forum for Agricultural Research in Africa (FARA) led by the executive director Dr. Yemi Akinbami are also acknowledged for sponsoring one of the authors for MSc. in UNIFEI, Brazil. Prof. Vladimir Oliveira Di Iorio of UFV is also acknowledged for coordinating the ARIFA programme in Brazil.

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